The Fock commutator of the CT generator

$[\hat{F}, \hat{A}]$, its two blocks, and the coefficients C^q_p, C^{rs}_{pq}

This is the derivation as it is written out by hand, with the algebra checked against explicit matrices. The convention is the one used there and in `formula.tex`: the Fock tensor carries its indices in the order f^p_q, the generator carries the *excitation* operator, and

$$\hat{F} = f^p_q \, \hat{E}^q_p, \qquad \hat{T}_1 = t^a_i \, \hat{E}^a_i, \qquad \hat{T}_2 = \tfrac{1}{2} \, t^{a\beta}_{ij} \, \hat{E}^{a\beta}_{ij}, \tag{1}$$

with i, j occupied, a, β virtual and $t^{a\beta}_{ij}$ antisymmetric in (a, β) and in (i, j). $\hat{A} = \hat{T} - \hat{T}^\dagger$ is anti-Hermitian and $\hat{T} = \hat{T}_1 + \hat{T}_2$.

Index ranges. Throughout, the *general* indices p, q, r, s run over the *whole* space, which is the union of the occupied and the virtual sets,

$$p, q, r, s \in \{i\} \cup \{a\}, \tag{2}$$

so a general index is an occupied one or a virtual one and nothing else. The $\hat{E}$ operators in the general basis, $\hat{E}^q_p$, therefore carry four families of blocks — occupied–occupied, occupied–virtual, virtual–occupied and virtual–virtual — and a Kronecker delta is what selects one of them. This is the convention that makes the enlargement of §4 work: the general range contains both the occupied and the virtual parts, so a restricted sum can be extended to it and then cut back with a delta.

1. The reduction

$$[\hat{F}, \hat{A}] = [\hat{F}, \hat{T}] + [\hat{F}, \hat{T}]^\dagger = [\hat{F}, \hat{T}_1] + [\hat{F}, \hat{T}_2] + \text{h.c.}, \tag{3}$$

because $\hat{A} = \hat{T} - \hat{T}^\dagger$ and $([\hat{F}, \hat{T}])^\dagger = -[\hat{F}, \hat{T}^\dagger]$. Only the two $\hat{T}$ blocks have to be computed; the other half of $\hat{C} = [\hat{F}, \hat{A}]$ is their conjugate.

2. $[\hat{F}, \hat{T}_1]$

Writing the operators out,

$$[\hat{F}, \hat{T}_1] = [f^p_q \hat{E}^q_p, \, t^a_i \hat{E}^a_i] = f^p_q \, t^a_i \, [\hat{E}^q_p, \hat{E}^a_i], \tag{4}$$

and with the $(1, 1)$ rule

$$[\hat{E}^q_p, \hat{E}^s_r] = \delta^s_p \, \hat{E}^q_r - \delta^q_r \, \hat{E}^s_p, \tag{5}$$

$$[\hat{E}^q_p, \hat{E}^a_i] = \delta^a_p \, \hat{E}^q_i - \delta^q_i \, \hat{E}^a_p. \tag{6}$$

The two deltas contract one index of f in each term — in the first the *lower* index of f becomes the virtual one, in the second the *upper* index becomes the occupied one — so

$$\boxed{[\hat{F}, \hat{T}_1] = t^a_i (f^a_q \, \hat{E}^q_i - f^p_i \, \hat{E}^a_p).} \tag{7}$$

Canonical Fock. With $f^p_q = \epsilon_p \delta^p_q$,

$$[\hat{F}, \hat{T}_1] = t^a_i \, (\epsilon_a - \epsilon_i) \, \hat{E}^a_i, \tag{8}$$

and the condition on the occupied–virtual block, $f^i_a + [\hat{F}, \hat{T}_1]^i_a = 0$, gives the singles amplitude

$$t^a_i = \frac{f^i_a}{\epsilon_i - \epsilon_a}, \tag{9}$$

the amplitude of the handwritten singles note.

3. $[\hat{F}, \hat{T}_2]$

In the same way,

$$[\hat{F}, \hat{T}_2] = \tfrac{1}{2} \, f^p_q \, t^{a\beta}_{ij} [\hat{E}^q_p, \hat{E}^{a\beta}_{ij}], \tag{10}$$

the $\frac{1}{2}$ being the one in $\hat{T}_2$ of (1). with the $(1, 2)$ rule

$$[\hat{E}^q_p, \hat{E}^{tu}_{rs}] = \delta^t_p \, \hat{E}^{qu}_{rs} + \delta^u_p \, \hat{E}^{qt}_{sr} - \delta^q_r \, \hat{E}^{tu}_{ps} - \delta^q_s \, \hat{E}^{ut}_{pr}, \tag{11}$$

so

$$[\hat{E}^q_p, \hat{E}^{a\beta}_{ij}] = \delta^a_p \, \hat{E}^{q\beta}_{ij} + \delta^\beta_p \, \hat{E}^{qa}_{ji} - \delta^q_i \, \hat{E}^{a\beta}_{pj} - \delta^q_j \, \hat{E}^{\beta a}_{pi},$$

so that

$$[\hat{F}, \hat{T}_2] = \tfrac{1}{2} \, f^p_q \, t^{a\beta}_{ij} \Big(\delta^a_p \, \hat{E}^{q\beta}_{ij} + \delta^\beta_p \, \hat{E}^{qa}_{ji} - \delta^q_i \, \hat{E}^{a\beta}_{pj} - \delta^q_j \, \hat{E}^{\beta a}_{pi} \Big), \tag{12}$$

where the $\frac{1}{2}$ is the one from $\hat{T}_2$ and must not be dropped: it is these four terms that are collected in §4, so the factor stands in front of the sum. Each delta contracts one index of f with the operator, which gives the block in the form

$$\boxed{\begin{aligned}[\hat{F}, \hat{T}_2] = \tfrac{1}{2} \Big(&+ f^a_q \, t^{a\beta}_{ij} \, \hat{E}^{q\beta}_{ij} + f^\beta_q \, t^{a\beta}_{ij} \, \hat{E}^{qa}_{ji} \\ &- f^i_p \, t^{a\beta}_{ij} \, \hat{E}^{a\beta}_{pj} - f^j_p \, t^{a\beta}_{ij} \, \hat{E}^{\beta a}_{pi} \Big). \end{aligned}} \tag{13}$$

The first and the second term are related by the antisymmetry of t (relabelling $a \leftrightarrow \beta$ and $i \leftrightarrow j$), and so are the third and the fourth. This relabelling relates the two *sums*; it is not available term by term, and term by term is how the block is used. At fixed values of the summation indices the four terms are different operators with different index patterns — $\hat{E}^{q\beta}_{ij}$ and $\hat{E}^{qa}_{ji}$ in the first pair, $\hat{E}^{a\beta}_{pj}$ and $\hat{E}^{\beta a}_{pi}$ in the second — so each of them restricts the general range to a different block, and each carries its own coefficient. Nothing is cancelled here: cancelling a pair would merge the two coefficients into one and lose the index patterns that §4 has to read C^{rs}_{pq} off.

Canonical Fock. With $f^p_q = \epsilon_p \delta^p_q$ every one of the four terms of (13) collapses onto $\hat{E}^{a\beta}_{ij}$, and the sum of the four gives

$$[\hat{F}, \hat{T}_2] = \tfrac{1}{2} \, t^{a\beta}_{ij} (\epsilon_a + \epsilon_\beta - \epsilon_i - \epsilon_j) \hat{E}^{a\beta}_{ij}. \tag{14}$$

Each of the four contributes one of the four orbital energies, which is what the uncollected form makes visible and what the $\frac{1}{2}$ in front of the sum is doing there.

4. Collecting into C_p^q and C_{pq}^{rs}

The result of §2 and §3 is a sum of $\hat{E}$ operators whose indices are partly restricted — $\hat{E}_i^q$ has an occupied lower index, $\hat{E}_p^a$ a virtual upper one. To read off coefficients in the general basis

$$[\hat{F}, \hat{A}] = C_p^q \, \hat{E}_p^q + C_{pq}^{rs} \, \hat{E}_{pq}^{rs}, \tag{15}$$

each restricted sum is enlarged to the general range (2) and restricted again with a delta. The enlargement is legitimate precisely because p, q cover the whole space: extending an occupied sum to all of $\{i\} \cup \{a\}$ adds the virtual labels, and the delta removes them again. For the one-body block,

$$\begin{aligned} \sum_{i,a,q} t_i^a f_q^a \, \hat{E}_i^q &= \sum_{p,q,i,a} t_i^a f_q^a \, \delta_i^p \, \hat{E}_p^q, \\ -\sum_{i,a,p} t_i^a f_i^p \, \hat{E}_p^a &= -\sum_{p,q,i,a} t_i^a f_i^p \, \delta_q^a \, \hat{E}_p^q, \end{aligned} \tag{16}$$

so the $[\hat{F}, \hat{T}]$ half of the one-body coefficient is

$$C_p^{[T]\,q} = \sum_{i,a} \Big(t_i^a f_q^a \, \delta_i^p - t_i^a f_i^p \, \delta_q^a \Big). \tag{17}$$

The conjugate half contributes the same two structures once more with the operator's indices exchanged, so the complete one-body coefficient is

$$\boxed{C_p^q = \underbrace{f_q^a \, t_p^a}_{p \text{ occupied}} - \underbrace{f_i^p \, t_i^q}_{q \text{ virtual}} + \underbrace{f_p^a \, t_q^a}_{q \text{ occupied}} - \underbrace{f_i^q \, t_i^p}_{p \text{ virtual}}} \tag{18}$$

two terms from $[\hat{F}, \hat{T}_1]$ and two from its conjugate.

The two-body coefficient is collected the same way, from the four terms of $[\hat{F}, \hat{T}_2]$ in (13) and the four of its conjugate, each on its own index pattern: $\hat{E}_{ij}^{q\beta}$ and $\hat{E}_{q\beta}^{ij}$, $\hat{E}_{ji}^{qa}$ and $\hat{E}_{qa}^{ji}$, $\hat{E}_{pj}^{a\beta}$ and $\hat{E}_{a\beta}^{pj}$, $\hat{E}_{pi}^{\beta a}$ and $\hat{E}_{\beta a}^{pi}$, carrying the coefficients $\pm\frac{1}{2} ft$ of that equation. Nothing is cancelled between them, for the reason given in §3: the eight are different operators, and it is their eight patterns that build up C_{pq}^{rs}.

With both coefficients complete, the result of the derivation is

$$\boxed{[\hat{F}, \hat{A}] = C_p^q \, \hat{E}_p^q + C_{pq}^{rs} \, \hat{E}_{pq}^{rs}.} \tag{19}$$

The two deltas cut the general range back down: δ_i^p keeps the *occupied* part of it for the lower index, so the first term contributes only for occupied p; δ_q^a keeps the *virtual* part for the upper index, so the second contributes only for virtual q. Every coefficient below is therefore written over the whole space, with the index ranges carried by the deltas rather than by the summation limits — which is the point of the rewriting.

The same enlargement applies to the four terms of (12), each operator now carrying four restrictions, one per index, which is why C_{pq}^{rs} comes out as a sum of delta products rather than single deltas. Two cautionary points, both of which have already bitten this derivation:

- **Keep each delta on the index it restricts.** The enlargement fixes the index of $\hat{E}$, not the index of f: the first term of (17) is $f_q^a \delta_i^p$, and writing it as $f_q^i \delta_a^p$ instead is wrong while still agreeing for symmetric f.

- **Canonicalise the two-body operator before reading the coefficient.** Once p, q run over all orbitals the same $\hat{E}^{rs}_{pq}$ is reached from more than one term; fixing one index ordering first is what prevents C^{rs}_{pq} and C^{sr}_{qp} being counted separately.

These coefficient tensors are the $\bar{h}$ and $\bar{g}$ of the CT Hamiltonian once the conjugate half of (3) is added.

5. What is verified

`tests/test_F_A_commutator.py` builds $\hat{F}, \hat{T}_1, \hat{T}_2$ as explicit spin-free matrices on four orbitals ($\{0,1\}$ occupied, $\{2,3\}$ virtual) with random tensors, and compares the closed forms against the matrix commutators:

quantity	max$\lvert\Delta\rvert$	relative
$[\hat{F}, \hat{T}_1]$, (7)	1.1×10^{-15}	2.6×10^{-16}
$[\hat{F}, \hat{T}_2]$, (13), the four terms uncancelled	3.6×10^{-15}	1.5×10^{-16}
the same block with the pairs folded	3.6×10^{-15}	1.5×10^{-16}
C^q_p of (18), one-body part	—	2.5×10^{-16}
(19) against $[\hat{F}, \hat{A}]$	4.4×10^{-15}	6.5×10^{-16}

The third and fourth rows are the two listings of the same operator: leaving the four terms uncancelled and folding the pairs agree to the same precision, which is why the note can keep them separate without changing anything.

The result is verified. `tests/test_F_A_result.py` takes the two blocks in closed form, adds the conjugate half, and collects every term into the general basis by running each term over its own index ranges — the enlarge-and-restrict step done mechanically, so that no delta has to be transcribed by hand — then reconstructs $C^q_p \hat{E}^q_p + C^{rs}_{pq} \hat{E}^{rs}_{pq}$ and compares it with the matrix commutator $[\hat{F}, \hat{A}]$: 4.4×10^{-15} absolute, 6.5×10^{-16} relative. The compact four-term form (18) of the one-body coefficient is checked separately, 2.5×10^{-16}.

One convention point the check turned up, worth stating because it is easy to overlook: with $\hat{F} = f^p_q \hat{E}^q_p$ the operator is Hermitian only when f is symmetric (real), which is the physical case. The step $[\hat{F}, \hat{A}] = [\hat{F}, \hat{T}] + [\hat{F}, \hat{T}]^\dagger$ uses that Hermiticity, and it fails for a non-symmetric f — the check therefore symmetrises f and asserts that $\hat{F}$ is Hermitian before comparing.

6. The other Fock convention

The derivation above uses $\hat{F} = f^p_q \hat{E}^q_p$. With the aligned order $\hat{F} = f^q_p \hat{E}^q_p = f^q_p \hat{a}^\dagger_q \hat{a}_p$, in which $f^q_p = \langle p | \hat{f} | q \rangle$, the same derivation gives

$$[\hat{F}, \hat{T}_1] = t^a_i (f^p_a \hat{E}^p_i - f^i_p \hat{E}^a_p), \qquad [\hat{F}, \hat{T}_2] = t^{a\beta}_{ij} (f^p_a \hat{E}^{p\beta}_{ij} - f^i_p \hat{E}^{a\beta}_{pj}). \tag{20}$$

The two sets are related by $f \to f^\mathsf{T}$, so they agree whenever f is symmetric — in particular for the diagonal canonical Fock, which is where a first check is usually made. That is what makes the distinction easy to lose; `tests/test_F_A_commutator.py` checks both conventions and asserts that each closed form deviates by a relative amount of order one under the other one's $\hat{F}$.

References

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